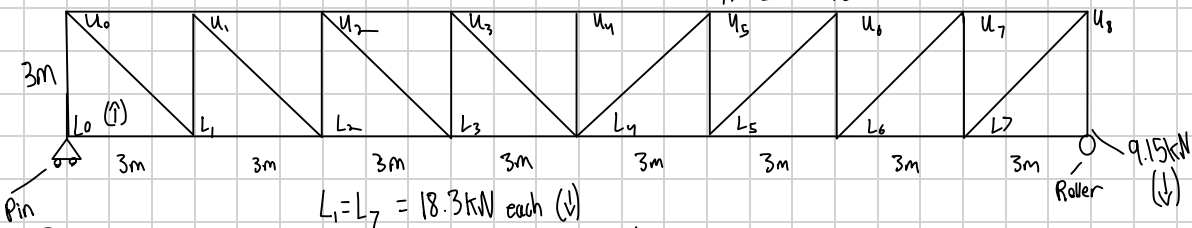
P2 Conveyor Gallery Truss

Sign Convention: Tension (+), Compression (-)

73.2 kN (↑) at L0 and L8

No loads U0-U8. Horizontal reaction L0 = 0



$L_1 = L_7 = 18.3$ kN each (↓)

Determinacy check

33 members + 3 reactions = 36 = 2x18 nodes.

Therefore the truss is determinate.

Assumed Loads

Dead: 1.5 kN/m conveyor + 0.5 kPa x 1 m grating = 2 kN/m

Live: 2.4 kPa x 1.0 m = 2.4 kN/m

ULS: 1.25x2 + 1.5x2.4 = 6.1 kN/m

Nodal loads: 18.3 kN down at L1-L7, and 9.15 kN at L0 and L8

Diagonal Length 4.243m, sin = cos = 0.7071

Reactions

Total load = 6.1 x 24 = 146.4 kN

Each reaction = $\frac{146.4}{2}$ = 73.2

Joint U0

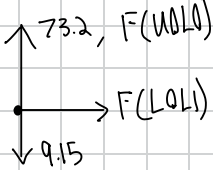


$\sum F_y = 0$: $64.05 - 0.7071 \cdot F(U0L1) = 0$, so $F(U0L1) = 90.58$ kN (T)

$\sum F_x = 0$: $F(U0U1) + 0.707(90.58) = 0$, so $F(U0U1) = -64.05$ kN

$= 64.05$ kN (C)

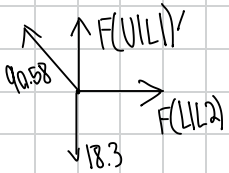
Joint L0



$\sum F_x = 0$: $F(L0L1) = 0$, so L0-L1=0 (zero force)

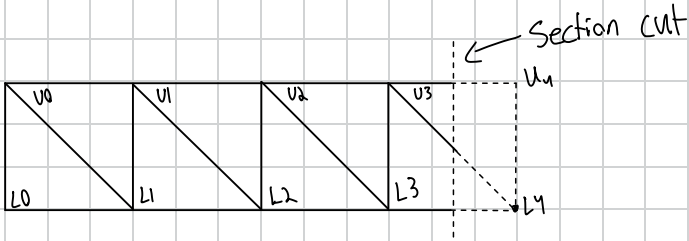
$\sum F_y = 0$: $73.2 - 9.15 + F(U0L0) = 0$, so $F(U0L0) = -64.05$, which is 64.05 kN (C)

Joint L1



$\sum F_y = 0: 0.7071(90.58) + F(U1L1) - 18.3 = 0$, So $F(U1L1) = -45.75 \rightarrow 45.75\,kN(C)$

$\sum F_x = 0: -0.7071(90.58) + F(L1L2) = 0$, So $F(L1L2) = 64.05\,kN(T)$



$\sum M_{L4} = 0: 73.2(12) - 9.15(12) - 18.3(9+6+3) - F(U3U4)\cdot(-3) = 0$

$F(U3U4) = \frac{-439.2}{3} = -146.40 \rightarrow 146.40\,kN(C)$

Full member force table (hand results above verified against robot model)

Group	Member	Mirror (right half)	Length (m)	Force (kN)	T / C
Top chord	U0–U1	U7–U8	3.000	−64.05	C
	U1–U2	U6–U7	3.000	−109.80	C
	U2–U3	U5–U6	3.000	−137.25	C
	U3–U4	U4–U5	3.000	−146.40	C
Bottom chord	L0–L1	L7–L8	3.000	0.00	0 (zero-force)
	L1–L2	L6–L7	3.000	+64.05	T
	L2–L3	L5–L6	3.000	+109.80	T
	L3–L4	L4–L5	3.000	+137.25	T
Vertical	U0–L0	U8–L8	3.000	−64.05	C
	U1–L1	U7–L7	3.000	−45.75	C
	U2–L2	U6–L6	3.000	−27.45	C
	U3–L3	U5–L5	3.000	−9.15	C
	U4–L4	—	3.000	0.00	0 (zero-force)
Diagonal	U0–L1	U8–L7	4.243	+90.58	T
	U1–L2	U7–L6	4.243	+64.70	T
	U2–L3	U6–L5	4.243	+38.82	T
	U3–L4	U5–L4	4.243	+12.94	T

AS-P2-CAL-002 Hand vs Robot member forces, Conveyor Gallery Truss (ULS 1.25D + 1.5L, self-weight excluded)

Prepared by: Abbas Saleh Date: 2026-10-18 Rev: 0 Student exercise, not for construction. Robot FX sign: + = compression; reversed below to match hand convention (+ = tension).

Bar	Member	Hand (kN)	Robot FX (kN)	Robot, hand sign (kN)	Difference (kN)	% diff	Status	Hand source
1	L0–L1	0.00	0.00	0.00	0.00	0.0%	PASS	Joints L0/U0/L1
2	L1–L2	64.05	-64.05	64.05	0.00	0.0%	PASS	Joints L0/U0/L1
3	L2–L3	109.80	-109.80	109.80	0.00	0.0%	PASS	Robot (symmetry check)
4	L3–L4	137.25	-137.25	137.25	0.00	0.0%	PASS	Robot (symmetry check)
5	L4–L5	137.25	-137.25	137.25	0.00	0.0%	PASS	Robot (symmetry check)
6	L5–L6	109.80	-109.80	109.80	0.00	0.0%	PASS	Robot (symmetry check)
7	L6–L7	64.05	-64.05	64.05	0.00	0.0%	PASS	Robot (symmetry check)
8	L7–L8	0.00	0.00	0.00	0.00	0.0%	PASS	Robot (symmetry check)
9	U0–U1	-64.05	64.05	-64.05	0.00	0.0%	PASS	Joints L0/U0/L1
10	U1–U2	-109.80	109.80	-109.80	0.00	0.0%	PASS	Robot (symmetry check)
11	U2–U3	-137.25	137.25	-137.25	0.00	0.0%	PASS	Robot (symmetry check)
12	U3–U4	-146.40	146.40	-146.40	0.00	0.0%	PASS	Section cut, panel 4
13	U4–U5	-146.40	146.40	-146.40	0.00	0.0%	PASS	Robot (symmetry check)
14	U5–U6	-137.25	137.25	-137.25	0.00	0.0%	PASS	Robot (symmetry check)
15	U6–U7	-109.80	109.80	-109.80	0.00	0.0%	PASS	Robot (symmetry check)
16	U7–U8	-64.05	64.05	-64.05	0.00	0.0%	PASS	Robot (symmetry check)
17	U0–L0	-64.05	64.05	-64.05	0.00	0.0%	PASS	Joints L0/U0/L1
18	U1–L1	-45.75	45.75	-45.75	0.00	0.0%	PASS	Joints L0/U0/L1
19	U2–L2	-27.45	27.45	-27.45	0.00	0.0%	PASS	Robot (symmetry check)
20	U3–L3	-9.15	9.15	-9.15	0.00	0.0%	PASS	Robot (symmetry check)
21	U4–L4	0.00	0.00	0.00	0.00	0.0%	PASS	Robot (symmetry check)
22	U5–L5	-9.15	9.15	-9.15	0.00	0.0%	PASS	Robot (symmetry check)
23	U6–L6	-27.45	27.45	-27.45	0.00	0.0%	PASS	Robot (symmetry check)
24	U7–L7	-45.75	45.75	-45.75	0.00	0.0%	PASS	Robot (symmetry check)
25	U8–L8	-64.05	64.05	-64.05	0.00	0.0%	PASS	Robot (symmetry check)
26	U0–L1	90.58	-90.58	90.58	0.00	0.0%	PASS	Joints L0/U0/L1
27	U1–L2	64.70	-64.70	64.70	0.00	0.0%	PASS	Robot (symmetry check)
28	U2–L3	38.82	-38.82	38.82	0.00	0.0%	PASS	Robot (symmetry check)
29	U3–L4	12.94	-12.94	12.94	0.00	0.0%	PASS	Robot (symmetry check)
30	U5–L4	12.94	-12.94	12.94	0.00	0.0%	PASS	Robot (symmetry check)
31	U6–L5	38.82	-38.82	38.82	0.00	0.0%	PASS	Robot (symmetry check)
32	U7–L6	64.70	-64.70	64.70	0.00	0.0%	PASS	Robot (symmetry check)
33	U8–L7	90.58	-90.58	90.58	0.00	0.0%	PASS	Robot (symmetry check)

Reactions: hand 73.20 kN each; Robot 73.20 kN each (sum 146.40 kN). Max force 146.40 kN = M/d check.

Members with hand source 'Robot (symmetry check)' were not individually hand-solved; they satisfy symmetry and joint U4 equilibrium.

AS-P2-MTO-001 Material Take-Off: Conveyor Gallery Truss Verification

Prepared by: Abbas Saleh Date: 2026-10-18 Rev: 0 Based on S-301 Rev 0; trial sections; centreline lengths. Student exercise, not for construction.

Mark	Member	Section (CISC)	Qty	Length each (m)	Total length (m)	Mass/m (kg/m)	Mass (kg)	Notes
TC	Top chord	HSS102x102x6.4	8	3.000	24.00	18.20	436.8	U0-U1 to U7-U8
BC	Bottom chord	HSS102x102x6.4	8	3.000	24.00	18.20	436.8	L0-L1 to L7-L8
V	Vertical	L76x76x6.4	9	3.000	27.00	7.28	196.6	U0-L0 to U8-L8
D	Diagonal	L76x76x6.4	8	4.243	33.94	7.28	247.1	U0-L1 to U3-L4, U5-L4 to U8-L7
Subtotal, one truss			**33**		**108.94**		**1317.3**	
Connection allowance	Gusset plates, bolts, welds					10%	131.7	10% as in AS-P1-MTO-001; replace once connections designed
TOTAL, ONE TRUSS							**1449.0**	**kg**
TOTAL, TWO TRUSSES			**2**				**2898.0**	**kg (assumed one truss each side of gallery)**

Checks

Item	Value	Target	Status
Member count	33	33	OK - matches 2j - r = 33
Chord length, each chord (m)	24.00	24	OK - equals span
Vertical count	9	9	OK - 8 panels + 1
Diagonal length (m)	4.243	4.243	OK - 3 m panel x 3 m depth

Mass per metre = A x 7850 kg/m3, with A from CISC section properties (HSS102x102x6.4: 2320 mm2; L76x76x6.4: 927 mm2), same method as AS-P1-MTO-001. Lengths are node-to-node centreline lengths from S-301; no allowance for cut lengths or end preparation.